

CHAPTER – 01

INTRODUCTION

1.1 – Project Overview

Electricity plays a vital role in modern life, powering homes, industries, businesses, and public services. Understanding electricity consumption patterns helps governments, utility providers, and consumers make informed decisions regarding energy management and conservation.

This project, "Plugging into the Future: An Exploration of Electricity Consumption Patterns," focuses on analysing electricity usage data to identify trends, consumption behaviours, and regional variations. Using SQL for data processing and Tableau for visualization, the project transforms raw electricity consumption data into meaningful insights.

The analysis examines electricity usage across different states, regions, metro cities, months, quarters, and years. It also compares consumption before and after lockdown periods, helping stakeholders understand how external events impact energy demand.

Objectives

- Analyse overall electricity consumption.
- Compare state-wise and region-wise electricity usage.
- Identify top and bottom consumers.
- Study monthly and yearly consumption trends.
- Evaluate electricity usage before and after lockdown.
- Explore metro city and state consumption patterns.
- Create interactive dashboards for better decision-making.

Tools Used

- SQL – Data cleaning, transformation, and analysis.
- Tableau – Data visualization and dashboard creation.
- Electricity Consumption Dataset – Source of consumption data.